

Windows Event Log Monitoring & Investigation

Final Project Report

Environment: Windows 11 Host + Wazuh Manager + Wazuh Agent

Tools: Windows Event Viewer, Wazuh, PowerShell, Command Prompt, VirtualBox

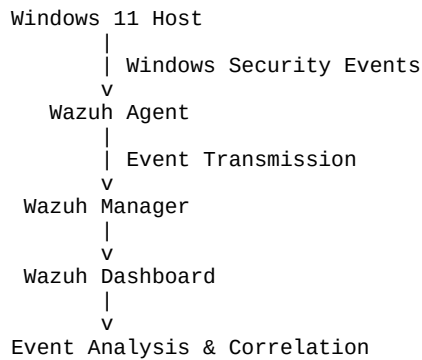
Project Purpose

This project demonstrates hands-on collection, monitoring, investigation, and correlation of Windows security events using Windows Event Logs and Wazuh. The work was performed in a controlled environment for educational and documentation purposes.

1. Project Overview

The project investigates Windows authentication, process creation, privilege assignment, and logoff activity. Windows Security Events were generated on the Windows endpoint, collected by the Wazuh Agent, transmitted to the Wazuh Manager, and examined through the Wazuh Dashboard.

Architecture



2. Objectives

- Understand Windows Security Event Logs
- Monitor Windows authentication activity
- Investigate failed and successful logons
- Analyze process creation events
- Capture process command-line activity
- Investigate special privilege assignments
- Correlate related Windows security events
- Use Wazuh to centralize and analyze Windows events
- Document investigation findings and evidence

3. Environment

- Windows 11 Host
- Ubuntu Wazuh Manager
- Wazuh Agent
- Wazuh Dashboard
- VirtualBox
- Windows Event Viewer
- PowerShell
- Command Prompt

4. Windows Events Investigated

Event ID 4624 — Successful Logon

Used to examine successful Windows authentication activity and identify the associated Logon ID.

Event ID 4625 — Failed Logon

A controlled failed authentication attempt was generated and investigated. The event included Logon Type 2, source address 127.0.0.1, and status 0xC000006D.

Event ID 4634 — Logoff

Used to identify when a Windows logon session ended and correlate the event using the Logon ID.

Event ID 4672 — Special Privileges Assigned

Used to investigate special privileges assigned to a Windows logon session, including privileges such as SeDebugPrivilege, SeBackupPrivilege, and SeRestorePrivilege.

Event ID 4688 — Process Creation

Used to investigate newly created processes, including process names, parent processes, process IDs, users, and command-line information.

5. Process Creation Investigation

Process Creation auditing was enabled on the Windows endpoint. Controlled commands including whoami and hostname were executed to generate Event ID 4688 activity.

- Command Line: hostname
- New Process: C:\Windows\System32\HOSTNAME.EXE
- Parent Process: C:\Windows\System32\cmd.exe
- Process ID: 0x3ca8
- User: vedha
- Wazuh Rule: 67027 — A process was created

6. Failed Logon Investigation

A temporary test account was used to generate a controlled incorrect-password attempt. Windows generated Event ID 4625 and Wazuh successfully detected the event.

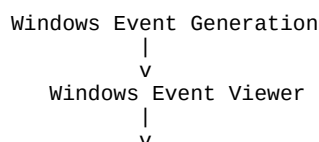
- Event ID: 4625
- Logon Type: 2 — Interactive
- Source IP: 127.0.0.1
- Status: 0xC000006D
- Wazuh Rule: 60122 — Logon Failure - Unknown user or bad password

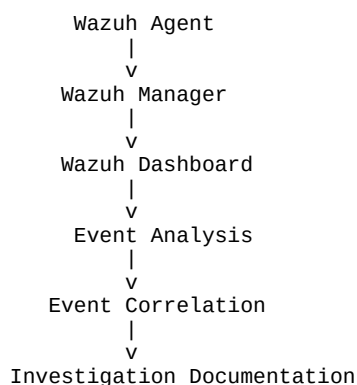
7. Privilege and Session Correlation

The Event ID 4672 investigation identified a special-privilege assignment with Subject Logon ID 0x26bfccb. A separate Event ID 4634 logoff event contained the same Target Logon ID, 0x26bfccb. This shared identifier was used to correlate the privilege-assignment and logoff events to the same Windows logon session.

The separately investigated Event ID 4624 screenshot contained a different Logon ID, 0x26c0ef0, so it was not treated as part of the specific 4672/4634 correlation.

8. Investigation Workflow





9. Evidence

The project contains 12 screenshots documenting Windows Event Viewer activity, Wazuh detection, event details, process activity, privilege assignments, and event correlation.

- 01-windows-event-viewer-4625.png
- 02-windows-event-viewer-4688.png
- 03-wazuh-agent-active.png
- 04-wazuh-4625-results.png
- 05-wazuh-4625-details.png
- 06-wazuh-4688-results.png
- 07-wazuh-4688-process-details.png
- 08-wazuh-4688-command-line.png
- 09-wazuh-4672-details.png
- 10-wazuh-4624-successful-logon.png
- 11-wazuh-4634-logoff.png
- 12-wazuh-4672-logon-correlation.png

10. Investigation Documentation

- 4624-successful-logon.md
- 4625-failed-logon.md
- 4634-logoff.md
- 4672-special-privileges.md
- 4688-process-creation.md

11. Repository Structure

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Windows-Event-Log-Monitoring-Investigation/
|
|■■■■ investigations/
|  ■■■■ 4624-successful-logon.md
|  ■■■■ 4625-failed-logon.md
|  ■■■■ 4634-logoff.md
|  ■■■■ 4672-special-privileges.md
|  ■■■■ 4688-process-creation.md
|
|■■■■ screenshots/
|  ■■■■ 01-windows-event-viewer-4625.png
|  ■■■■ 02-windows-event-viewer-4688.png
|  ■■■■ 03-wazuh-agent-active.png
|  ■■■■ 04-wazuh-4625-results.png
|  ■■■■ 05-wazuh-4625-details.png
|  ■■■■ 06-wazuh-4688-results.png
|  ■■■■ 07-wazuh-4688-process-details.png
|  ■■■■ 08-wazuh-4688-command-line.png
  
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| ■■■ 09-wazuh-4672-details.png
| ■■■ 10-wazuh-4624-successful-logon.png
| ■■■ 11-wazuh-4634-logoff.png
| ■■■ 12-wazuh-4672-logon-correlation.png
■■■ README.md

12. Project Outcome

This project provided practical experience with Windows security event monitoring, Wazuh-based event collection, process monitoring, authentication analysis, privilege monitoring, and event correlation.

The project demonstrates a practical workflow for investigating Windows security events from event generation through centralized monitoring, analysis, correlation, and documentation.

13. Conclusion

The investigation successfully demonstrated collection and analysis of Windows Security Events 4624, 4625, 4634, 4672, and 4688 through Wazuh. The work also demonstrated how Logon IDs can be used to correlate related events and how process creation data can provide useful endpoint investigation context.

Disclaimer

All security events and authentication activities in this project were generated and investigated in a controlled environment for educational and documentation purposes.